

United International University
Department of Computer Science and Engineering



Final Examination
Course Code: CSE 1110
Course Title: Introduction to Computer System
Section: E Set: B

Total Marks: 20

Duration: 1 Hour

Q1. (Conditions – 10 Marks)

Chernobyl Reactor Safety Protocol (Using if-else)

The Chernobyl Power Plant has various **reactor modes** and corresponding **alert levels**. Write a C program that takes two inputs:

1. Reactor mode (1–3)
2. Alert level (1–3)

Use a **nested switch-case** structure to determine the response protocol as follows:

- **Case 1: Normal Operation**
 - Alert 1: Print “System stable.”
 - Alert 2: Print “Minor fluctuation detected.”
 - Alert 3: Print “Begin diagnostic scan.”
- **Case 2: Cooling Operation**
 - Alert 1: Print “Coolant pressure optimal.”
 - Alert 2: Print “Coolant temperature rising.”
 - Alert 3: Print “Emergency coolant release activated.”
- **Case 3: Containment Mode**
 - Alert 1: Print “Containment seals secure.”
 - Alert 2: Print “Minor radiation leak detected.”
 - Alert 3: Print “Major breach! Evacuate control room!”
- **Default:** Print “Invalid reactor mode.”

Sample Input:

Enter reactor mode: 3

Enter alert level: 2

Sample Output:

Mode: Containment

Alert Level: 2

Status: Minor radiation leak detected.

Q2. (Loop – 10 Marks)**Prime Check and Factor Sum for Reactor ID**

Each reactor in the Chernobyl plant has a numeric ID. Write a C program to perform the following:

1. Input a reactor ID number.
2. Check if the reactor ID is a prime number.
3. If it is **prime**, display “Prime”.
4. If it is **not prime**, display “Not Prime” and also calculate the sum of all its factors excluding itself.

Constraints: Use a single ‘for’ or ‘while’ loop; no nested loops.

Sample Input 1:

Enter Reactor ID: 13

Sample Output 1:

Reactor ID: 13

Prime

Sample Input 2:

Enter Reactor ID: 12

Sample Output 2:

Reactor ID: 12

Not Prime

Sum of Factors: 28

End of Question Paper